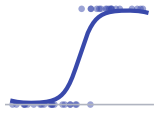
DATA

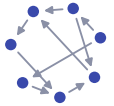
Track 1 · Synthetic 1-D

Known posterior $\eta(x)$
Legendre basis models



Track 2 · SERGIO

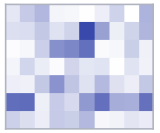
Simulated scRNA-seq, known GRN
per-(cell, pair) features, cell split



i.i.d. by construction

Track 3 · DREAM5

Real benchmark
empirical case study



TRAINING & CALIBRATION

Probabilistic classifier

$f_\theta(x) = \hat{p}$ (edge probability)

End-to-end calibration

$$\min_\theta \mathcal{L}_{\text{base}} + \lambda \widehat{\text{CE}}$$

base: CE, Brier
$\widehat{\text{CE}}$: Soft-ECE, MMCE, ECE_{KDE}

Baselines

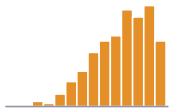
Post-hoc: temperature, Platt, isotonic, Beta
Training: focal loss

Hyperparameter sweep

λ grid, capacity m

Predicted probabilities

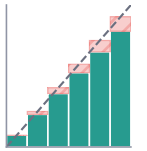
$\hat{p}$ per test sample



EVALUATION

Calibration

ECE, reliability diagrams

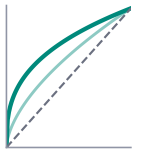


Calibration–refinement

risk = calibration + refinement

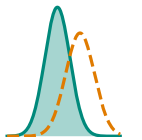
Discrimination

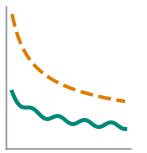
AUROC, AUPR, Early Precision



Robustness

RobustnessGap under shift



Generalisation bounds

PAC-Bayes, Rademacher
(Tracks 1–2 only)